

Stochastic Processes

Random Variable. $X: \Omega \longrightarrow \mathbb{R}$. $F_X(x)$, $f_X(x)$, $E(X^k)$

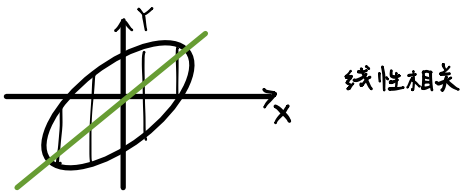
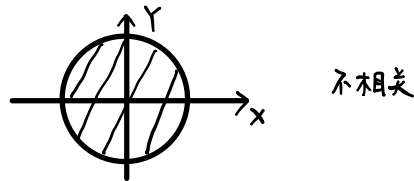
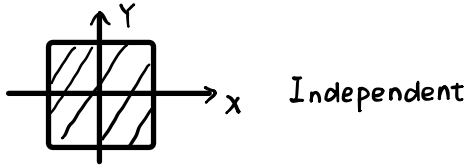
在 Stochastic Processes 中为 Multiple R.V.

Relation

① Linear Correction	$\begin{cases} \text{Time} \\ \text{Frequency} \\ \text{Gaussian} \end{cases}$	② Markov Property	$\begin{cases} \text{Discrete Time} \\ \text{Continuous Time} \\ \text{Poisson} \end{cases}$
③ Martingale (鞅)			

Linear Correction

Binary Relation. X, Y Joint Distribution, $f_{X,Y}(x,y)$



$Y \sim kX$ 寻找两者之间的关系. Metric (Distance): $(E|X-Y|^2)^{\frac{1}{2}}$
 $E|Y-kX|^2 = E|Y|^2 + k^2 E|X|^2 - 2kE(XY)$ Correlation

① Independent $\longleftrightarrow$ Uncorrelated (Orthogonal) $E(XY) = 0$

e.g. 不相关却不独立的例子

$\theta \sim U(0, 2\pi)$, 定义 $X = \cos(\theta)$, $Y = \sin(\theta)$

$$X^2 + Y^2 = 1$$

$$E(X) = \int_{-\infty}^{+\infty} \cos(\theta) f_{\theta}(\theta) d\theta = \int_0^{2\pi} \cos(\theta) \cdot \frac{1}{2\pi} d\theta = 0$$

$$\text{同理: } E(Y) = 0$$

$$\text{而 } E(XY) = \int_0^{2\pi} \cos(\theta) \sin(\theta) \cdot \frac{1}{2\pi} d\theta = 0$$

$$\text{故 } E(XY) = E(X)E(Y) = 0 \longrightarrow \text{不相关}$$

② $|E(XY)| \leq (E(X^2)E(Y^2))^{\frac{1}{2}}$ Cauchy Inequality $(\sum_k x_k y_k) \leq (\sum_k x_k^2 \sum_k y_k^2)^{\frac{1}{2}}$

$$\text{内积: } |\langle X, Y \rangle| \leq (\langle X, X \rangle \langle Y, Y \rangle)^{\frac{1}{2}}$$

$$\text{Pf: } g(\alpha) = \langle \alpha X + Y, \alpha X + Y \rangle = \alpha^2 \langle X, X \rangle + 2\alpha \langle X, Y \rangle + \langle Y, Y \rangle \geq 0$$

$$\text{故 } \Delta = \frac{4\langle X, Y \rangle^2 - 4\langle X, X \rangle \langle Y, Y \rangle}{\langle X, X \rangle^2} \leq 0$$

$$\text{因此 } \langle X, Y \rangle^2 \leq \langle X, X \rangle \langle Y, Y \rangle$$

$$\text{从而 } |\langle X, Y \rangle| \leq (\langle X, X \rangle \langle Y, Y \rangle)^{\frac{1}{2}}$$

□

$$\textcircled{3} (x_1, \dots, x_n)^T = X$$

$$\text{Correlation Matrix: } E(X X^T) = R_X, \quad R_X(i, j) = E(X_i X_j)$$

$$1. \text{ 对称性: } R_X(i, j) = E(X_i X_j) = E(X_j X_i) = R_X(j, i)$$

$$\text{则: } R_X = R_X^T$$

$$2. \text{ 正定性: } \forall z \in \mathbb{R}^n, \quad z^T R_X z = z^T E(X X^T) z = E(z^T X X^T z) \\ = E(X^T z)^2 \geq 0$$

Def: (Stochastic Processes)

$X(t)$. Random Function 更严格的写法: $X(W, t)$, $W \in \Omega$

给定 W , 为 Sample Path (样本轨)

$$R_X(t, s) = E(X(t) X(s)) \longrightarrow (\text{Auto}) \text{ Correlation Function}$$

相关函数的性质: ① $R_X(t, s) = R_X(s, t)$

$$\textcircled{2} R_X(t, s) \geq 0$$

$$\textcircled{3} |R_X(t, s)| \leq (R_X(t, t) R_X(s, s))^{\frac{1}{2}}$$

Stationary (Invariance) - 共四种

Wide Sence, w.s.s : ① $E(X(t)) \equiv m$.

$$\textcircled{2} R_X(t+T, s+T) = R_X(t, s), \quad \forall T$$

此时, R_X 的性质为: ① $R_X(-\tau) = R_X(\tau)$

$$\textcircled{2} R_X(0) \geq 0$$

$$\textcircled{3} |R_X(\tau)| \leq R_X(0)$$

e.g. Modulated Signal

$$X(t) = A(t) \cos(2\pi f t + \theta), \quad A(t), \theta, \text{ random. } A(t) \text{ w.s.s. independent}$$

$$\textcircled{1} E(X(t)) = E(A(t) \cos(2\pi f t + \theta)) = E(A(t)) E(\cos(2\pi f t + \theta))$$

$$\text{而 } E(\cos(2\pi f t + \theta)) = \frac{1}{2\pi} \int_0^{2\pi} \cos(2\pi f t + \theta) d\theta = 0$$

$$\text{故 } E(X(t)) = 0$$

$$\textcircled{2} E(A(t) \cos(2\pi f t + \theta) A(s) \cos(2\pi f s + \theta))$$

$$= E(A(t) A(s)) E(\cos(2\pi f t + \theta) \cos(2\pi f s + \theta))$$

$$\text{其中 } E(A(t) A(s)) = R_A(t-s)$$

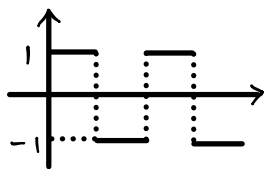
$$\text{而 } E(\cos(2\pi f t + \theta) \cos(2\pi f s + \theta)) = \frac{1}{2\pi} \int_0^{2\pi} \cos(2\pi f t + \theta) \cos(2\pi f s + \theta) d\theta$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \left[\frac{1}{2} \cos(2\pi f (t-s)) + \frac{1}{2} \cos(2\pi f (t+s) + 2\theta) \right] d\theta$$

$$= \frac{1}{2} \cos(2\pi f (t-s))$$

$$\text{故 } E(A(t) \cos(2\pi f t + \theta) A(s) \cos(2\pi f s + \theta)) = R_A(t-s) \cdot \frac{1}{2} \cos(2\pi f (t-s)).$$

e.g. Random Telegraph Signal $X(t)$



单位时间内跳跃: $N(s) \sim P(\lambda s)$

$$P(N(s)=k) = \frac{(\lambda s)^k}{k!} \exp(-\lambda s)$$

$$X(0) \sim \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \end{pmatrix}$$

$$\textcircled{1} E(X(t)) = \sum_k k P(X(t)=k) = 1 \cdot P(X(t)=1) + (-1) \cdot P(X(t)=-1)$$

$$P(X(t)=1) = \underbrace{P(X(t)=1 | X(0)=-1)} P(X(0)=-1) + P(X(t)=1 | X(0)=1) P(X(0)=1)$$

$N(t)$ 为奇数

$$P(N(t) \text{ is odd}) = \sum_{k \text{ odd}} \frac{(\lambda t)^k}{k!} \exp(-\lambda t)$$

$$\text{考虑 } \exp(\lambda t) = \sum_k \frac{(\lambda t)^k}{k!}, \quad \exp(-\lambda t) = \sum_k \frac{(-\lambda t)^k}{k!} = \sum_k (-1)^k \frac{(\lambda t)^k}{k!}$$

$$\begin{aligned} \text{故 } P(N(t) \text{ is odd}) &= \frac{1}{2} (\exp(\lambda t) - \exp(-\lambda t)) \cdot \exp(-\lambda t) \\ &= \frac{1}{2} (1 - \exp(-2\lambda t)) \end{aligned}$$

$$P(N(t) \text{ is even}) = 1 - P(N(t) \text{ is odd}) = \frac{1}{2} (1 + \exp(-2\lambda t))$$

$$\begin{aligned} \text{因此 } P(X(t)=1) &= \frac{1}{4} (1 + \exp(-2\lambda t)) + \frac{1}{4} (1 - \exp(-2\lambda t)) \\ &= \frac{1}{2} \end{aligned}$$

$$\text{故 } E(X(t)) = 0$$

$$\textcircled{2} E(X(t)X(s)) = 1 \cdot \underbrace{P(X(t)X(s)=1)} + (-1) \cdot P(X(t)X(s)=-1)$$

在 $s \rightarrow t$ 时跳变了偶数次

$$\begin{aligned} &= 1 \cdot \frac{1}{2} [1 + \exp(-2\lambda(t-s))] + (-1) \cdot \frac{1}{2} [1 - \exp(-2\lambda(t-s))] \\ &= \exp(-2\lambda(t-s)) \end{aligned}$$

$$R_X(\tau) = \exp(-2\lambda|\tau|)$$

$$X(t) \text{ w.s.s } R_X(0) = R_X(T) \Rightarrow R_X(T+\tau) = R_X(\tau), \quad \forall \tau$$

$$\begin{aligned} \text{pf: } E|X(t) - X(t+T)|^2 &= E|X(t)|^2 + E|X(t+T)|^2 - 2E|X(t)X(t+T)| \\ &= R_X(0) + R_X(0) - 2R_X(T) \\ &= 2(R_X(0) - R_X(T)) \\ &= 0 \end{aligned}$$

$$\begin{aligned} |R_X(\tau) - R_X(\tau+T)| &= |E(X(0)X(\tau)) - E(X(0)X(\tau+T))| \\ &= |E[X(0)(X(\tau) - X(\tau+T))]| \\ &\leq E(|X(0)| \cdot |X(\tau) - X(\tau+T)|) \\ &\leq \left[E(X^2(0)) E(|X(\tau) - X(\tau+T)|^2) \right]^{\frac{1}{2}} \\ &= 0 \end{aligned}$$

$$\text{故 } R_X(\tau) = R_X(\tau+T)$$

□

Prop: $x(t)$, w.s.s. $R_x(z)$ 在零点连续, 则 $R_x(z)$ 在任何一点连续.

Positive Definite Functions:

$$f(x), \text{ p.d.} \Leftrightarrow \forall n, \forall \alpha_1, \dots, \alpha_n, (f(\alpha_i - \alpha_j))_{ij} \text{ 为 P.d.}$$

Prop: $R_x(z)$ 为一个正定函数.

pf: $R_x(i, j) = R_x(t_i - t_j) = E(x(t_i) x(t_j))$, $X = (x(t_1), \dots, x(t_n))^T$

$$R_x = E(X X^T) \geq 0$$

□

In fact, Correlation Function $\Leftrightarrow$ P.D. Function

Bochner: $f(x)$ is p.d. $\Leftrightarrow \int_{-\infty}^{+\infty} f(x) \exp(-j\omega x) dx \geq 0$

pf: 充分性 " $\Leftarrow$ "

$$F(\omega) = \int_{-\infty}^{+\infty} f(x) \exp(-j\omega x) dx \Rightarrow f(x) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\omega) \exp(j\omega x) d\omega$$

断言: ① $\exp(j\omega x)$ is p.d.

pf: $\forall n, \forall \alpha_1, \dots, \alpha_n, R = (\exp(j\omega(\alpha_i - \alpha_j)))_{ij}$

$$\forall z = (z_1, \dots, z_n)^T \in \mathbb{C}^n, z^H R z = \sum_{i=1}^n \sum_{j=1}^n R_{ij} z_i \bar{z}_j$$

$$= \sum_{i=1}^n \sum_{j=1}^n \exp(j\omega(\alpha_i - \alpha_j)) z_i \bar{z}_j$$

$$= \left(\sum_{i=1}^n \exp(j\omega \alpha_i) z_i \right) \left(\sum_{j=1}^n \exp(-j\omega \alpha_j) \bar{z}_j \right)$$

$$= \left(\sum_{i=1}^n \exp(j\omega \alpha_i) z_i \right) \overline{\left(\sum_{j=1}^n \exp(j\omega \alpha_j) z_j \right)}$$

$$= \left| \sum_{i=1}^n \exp(j\omega \alpha_i) z_i \right|^2 \geq 0 \quad \#$$

② $\forall f_1, \dots, f_n \geq 0, \sum_{i=1}^n f_i \exp(j\omega_i x)$ is p.d.

③ $\int_{-\infty}^{+\infty} F(\omega) \exp(j\omega x) d\omega = 2\pi f(x)$ is p.d.

必要性 " $\Rightarrow$ "

Fix n , choose $\alpha_1, \dots, \alpha_n, (f(\alpha_i - \alpha_j))_{ij}$ 取 $z = (\exp(j\omega \alpha_1), \dots, \exp(j\omega \alpha_n))$

$$0 \leq z^H R z = \sum_{i=1}^n \sum_{j=1}^n f(\alpha_i - \alpha_j) \exp(j\omega \alpha_i) \overline{\exp(j\omega \alpha_j)}$$

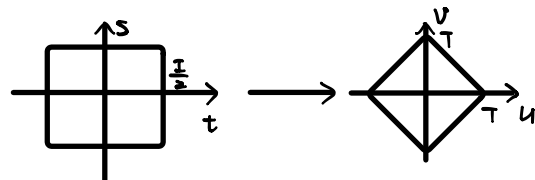
$$= \sum_{i=1}^n \sum_{j=1}^n f(\alpha_i - \alpha_j) \exp(j\omega(\alpha_i - \alpha_j))$$

$$\longrightarrow \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} \int_{-\frac{T}{2}}^{\frac{T}{2}} f(t-s) \exp(j\omega(t-s)) dt ds$$

令 $\begin{cases} u = t-s \\ v = t+s \end{cases} \quad = \frac{1}{T} \iint_{\diamond} f(u) \exp(j\omega u) \cdot \frac{1}{2} du dv$

$$= \frac{1}{2T} \int_{-T}^T \int_{|u|-T}^{-|u|+T} f(u) \exp(j\omega u) dv du$$

$$= \frac{1}{T} \int_{-T}^T (T - |u|) f(u) \exp(j\omega u) du$$



$$= \int_{-T}^T \left(1 - \frac{|u|}{T}\right) f(u) \exp(j\omega u) du$$

$$\xrightarrow{T \rightarrow \infty} \int_{-\infty}^{+\infty} f(u) \exp(j\omega u) du \geq 0$$

□